

3. Final exam:
Mathematics for Artificial Intelligence 2

Last name:

First name:

Matr. nr.:

Linz/Vienna/Bregenz, September 22nd, 2022

Signature:

Duration: 180 minutes

Maximum of 58 points

Rules:

- **Put your name and matriculation nr. on every sheet!**
 - **Use a new sheet for every exercise!**
 - Electronic devices are not allowed!
 - Do not use pencil or a red pen.
 - The whole procedure of solution is required.
 - All results must be simplified as far as possible.
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Good luck!

EXERCISE 1: **Curve Sketching** (20 points)

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) := \begin{cases} \frac{1}{x} \cdot e^{-\frac{1}{x^2}}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- a) Where is f continuous and differentiable, respectively?
- b) Determine all local and global extrema and extreme values of f .
- c) Where is f non-decreasing/non-increasing?
- d) Make a sketch of the graph of f .

EXERCISE 2: **Taylor series** (12 points)

Determine the Taylor polynomial $T_5(x)$ of order 5 of the function

$$f: (0, \infty) \rightarrow \mathbb{R}, \quad f(x) = x^2 \cdot \ln(x)$$

around $x_0 = 1$.

EXERCISE 3: **Antiderivatives** (8 points)

- a) Find one antiderivative of the following functions, respectively:

- $f_1(x) = (x + 1)e^x$
- $f_2(x) = \frac{x^2 + 2x + 2}{x^3 + 3x^2 + 6x + 6}$
- $f_3(x) = x^4 \cdot \cos(x^5 + \pi)$

- b) Find the function f on $\mathbb{R}$, which satisfies $f'(x) = f_3(x)$ for all $x \in \mathbb{R}$ and $f(0) = 42$.

EXERCISE 4: **Extrema under constraints** (12 points)

Calculate the global extrema of

$$f(x_1, x_2) = (x_1 + 4)^2 + (x_2 - 4)^2$$

under the constraint

$$x_1^2 + x_2^2 \leq 8.$$

(You do not have to prove that an extrema exists.)

EXERCISE 5: **Jacobian** (6 points)

We consider the function $F = f \circ g$, where

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad f(a, b) = (a^2, b^2, a^2 - b^2),$$

$$g: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad g(x, y) = (x^2 + y^2, 1 + \sin(x - y)).$$

Compute the Jacobian of F at the point $(x_0, y_0) = (1/2, 1/2)$.